

Two-Way Fixed Effect with Differential Timing and Goodman-Bacon Decomposition

Samuel Rowe adapted from Cunningham (2021)

2026-04-20

Contents

Overview	5
1 Two-way Fixed Effects with Differential Timing	7
1.1 Replicate Cheng and Hoesktra (2013)	7
1.2 Inspect the data	8
1.3 TWFE with Differential Timing	11
1.4 Event Study	16
2 Goodman-Bacon Decomposition	21
2.1 <i>estat bdecomp</i>	21
2.2 <i>bacondecomp</i>	24
2.3 <i>ddtiming</i>	26

bookdown::render_book()

Overview

Topics:

1. Two-Way Fixed Effects with Differential Timing
2. Goodman-Bacon Decomposition

Chapter 1

Two-way Fixed Effects with Differential Timing

1.1 Replicate Cheng and Hoekstra (2013)

We will replicate Cheng and Hoekstra (2013) in our Two-Way Fixed Effects (TWFE) example. Cheng and Hoekstra (2013) evaluate the impact of gun reform on violence with time differential adoption of gun reform after the death of Trayvon Martin in 2012.

The policy change of interest are castle-doctrine statutes (or “Stand Your Ground” laws). Between 2000 and 2010, 21 states expanded their castle-doctrine by extending the castle-doctrine to outside of the home where lethal force could legally be used. It eliminated the long-standing common law of a victim’s duty to retreat. And, if the victim felt threatened, then they could legally use lethal force.

The authors assess the reform of castle-doctrine statutes on homicides. Proponents argue that it reduces crime through deterrence, while opponents argue it increases homicides due to more guns being present. Cunningham argues that these doctrines reduce the marginal cost of manslaughter by reducing civil liability.

The authors utilize a TWFE estimator, since there are differential timings in treatment. States did not adopt castle-doctrine reform simultaneously. Therefore, we need to be concerned about comparing early treated to late treated.

Model

$$Y_{it} = \alpha + \delta D_{it} + \gamma X_{it} + \lambda_i + \tau_t + \varepsilon_{it}$$

Where

- Y_{it} is the homicide rate per 100k
- D_{it} is the binary for castle doctrine present in state i at time t
- X_{it} are region-by-region fixed effects (so a treatment state's counterfactual comes from its region)
- λ_i are state fixed effects
- τ_t are time fixed effects
- ε_{it} is the idiosyncratic error clustered at the state level

We have a few options for implementing Goodman-Bacon Decomposition. The first one is the *bacondecomp* package and the other is a post-estimation command with *xtddidregress*.

```
ssc install bacondecomp
```

1.2 Inspect the data

We will import our data and inspect the outcomes and treatment variables.

```
clear
cd "/Users/Sam/Desktop/Econ 672/Data"
use castle

sum homicide, detail
```

```
/Users/Sam/Desktop/Econ 672/Data
```

homicide count per 100,000 state population				

	Percentiles	Smallest		
1%	.9615808	.6466538		
5%	1.358292	.7829468		
10%	1.792752	.819343	Obs	550
25%	2.642913	.8362355	Sum of wgt.	550
50%	4.652019		Mean	4.761399
		Largest	Std. dev.	2.482834
75%	6.429322	13.10705		
90%	7.707187	13.39059	Variance	6.164465
95%	8.774975	13.63958	Skewness	.6841787
99%	12.92404	14.52561	Kurtosis	3.558378

Our median homicide rate per 100K is 4.65, while the mean is 4.76.

We'll inspect the histogram of both homicide per 100K and the natural log of homicides per 100K.

```
histogram homicide
```

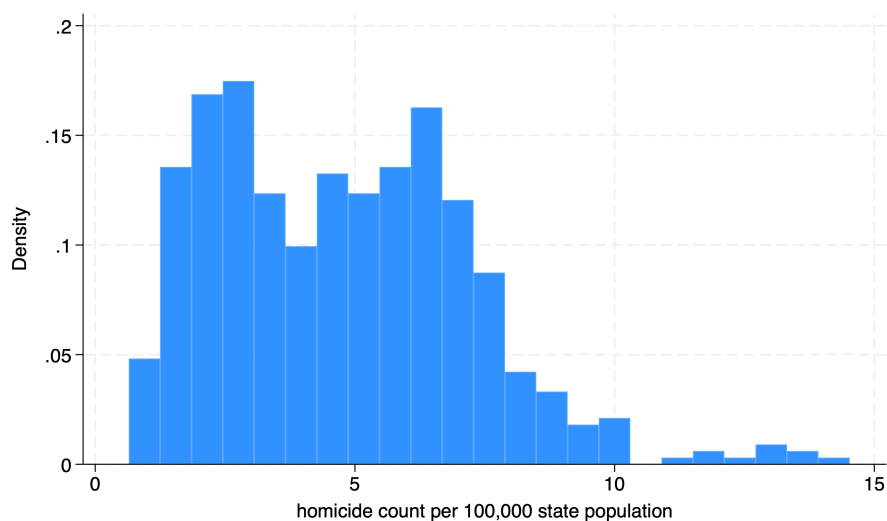


Figure 1.1: Homicides per 100K

```
histogram l_homicide, xtitle("Natural Log of Homicides per 100K")
```

Next, we will inspect our treatment variable of interest, which is the change in states adopting caste-doctrine statutes in state i at time t .

```
label variable post "Year of treatment"
tab year post
```

year	Year of treatment		Total
	0	1	
2000	50	0	50
2001	50	0	50
2002	50	0	50
2003	50	0	50
2004	50	0	50
2005	50	0	50
2006	49	1	50

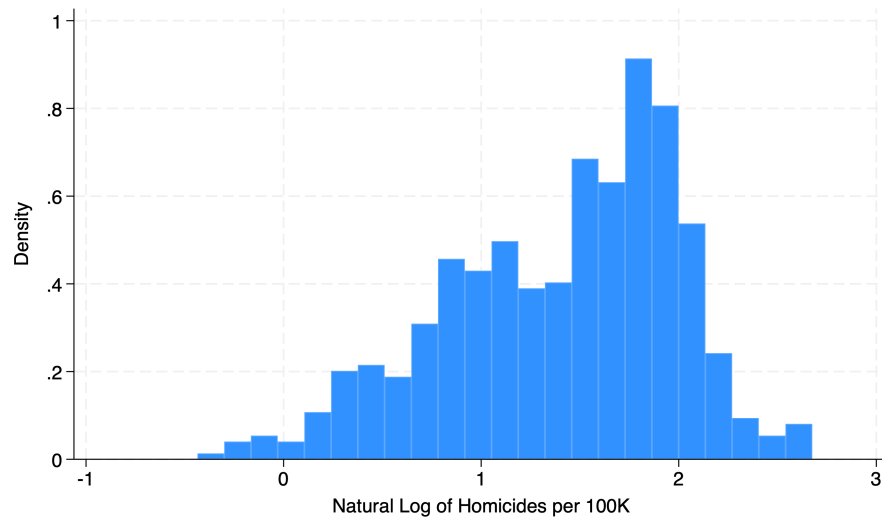


Figure 1.2: Natural Log of Homicides per 100K

2007		36	14		50
2008		32	18		50
2009		30	20		50
2010		29	21		50
Total		476	74		550

We can see that 1 state adopted the policy in 2006. There were 13 more adopters in 2007, 4 more in 2008, 2 more in 2009, and 1 more in 2010. The key issue is that states did not adopt castle-doctrine statutes simultaneously.

Next, we will set up our global macros first to replicate Cheng and Hoesktra (2013). Our state level covariates, X_{it} , includes demographic variables, state linear trends, region, LN of exogenous crime rates, LN of public expenditures, LN of FTE police, unemployment rates, poverty rates, LN of median state income, LN of incarceration rate, and a lag in LN of incarceration rate.

```
global crime1 jhccitizen_c jhpolice_c murder homicide robbery assault burglary larceny
global demo blackm_15_24 whitem_15_24 blackm_25_44 whitem_25_44 //demographics
global lintrend trend_1-trend_51 //state linear trend
global region r20001-r20104 //region-quarter fixed effects
global exocrime l_larceny l_motor // exogenous crime rates
global spending l_exp_subsidy l_exp_pubwelfare
global xvar l_police unemployrt poverty l_income l_prisoner l_lagprisoner $demo $spend.
```

1.3 TWFE with Differential Timing

We will use our *xtreg* command to estimate the TWFE estimator.

```
xtset sid year
xtreg l_homicide i.year $region $xvar $lintrend post [aweight=popwt], fe vce(cluster sid)
```

Panel variable: sid (strongly balanced)

Time variable: year, 2000 to 2010

Delta: 1 unit

note: r20004 omitted because of collinearity.
 note: r20014 omitted because of collinearity.
 note: r20024 omitted because of collinearity.
 note: r20034 omitted because of collinearity.
 note: r20044 omitted because of collinearity.
 note: r20054 omitted because of collinearity.
 note: r20064 omitted because of collinearity.
 note: r20074 omitted because of collinearity.
 note: r20084 omitted because of collinearity.
 note: r20094 omitted because of collinearity.
 note: r20101 omitted because of collinearity.
 note: r20102 omitted because of collinearity.
 note: r20103 omitted because of collinearity.
 note: r20104 omitted because of collinearity.
 note: trend_9 omitted because of collinearity.
 note: trend_46 omitted because of collinearity.
 note: trend_49 omitted because of collinearity.
 note: trend_50 omitted because of collinearity.
 note: trend_51 omitted because of collinearity.

Fixed-effects (within) regression

Group variable: sid

Number of obs = 550

Number of groups = 50

R-squared:

Within = 0.5851

Between = 0.3750

Overall = 0.3208

Obs per group:

min = 11

avg = 11.0

max = 11

corr(u_i, Xb) = -0.9808

F(49, 49)

Prob > F

=

=

.

.

(Std. err. adjusted for 50 clusters in sid)

|

Robust

12CHAPTER 1. TWO-WAY FIXED EFFECTS WITH DIFFERENTIAL TIMING

l_homicide	Coefficient	std. err.	t	P> t	[95% conf. interval]	
year						
2001	.0684684	.0364135	1.88	0.066	-.0047073	.141644
2002	.07167	.0436393	1.64	0.107	-.0160265	.1593665
2003	.1006137	.0449905	2.24	0.030	.0102019	.1910256
2004	.1293868	.0368288	3.51	0.001	.0553766	.203397
2005	.1605832	.0401576	4.00	0.000	.0798834	.2412829
2006	.1743047	.0444244	3.92	0.000	.0850304	.263579
2007	.0903824	.0452763	2.00	0.051	-.0006037	.1813685
2008	.0017728	.043145	0.04	0.967	-.0849304	.088476
2009	-.0830581	.0747786	-1.11	0.272	-.2333313	.067215
2010	-.1640143	.0898474	-1.83	0.074	-.3445694	.0165409
r20001	.0680535	.5230205	0.13	0.897	-.9829956	1.119103
r20002	.1947373	.0830414	2.35	0.023	.0278593	.3616152
r20003	-.7696014	.1386803	-5.55	0.000	-1.04829	-.490913
r20004	0	(omitted)				
r20011	.0245472	.4611295	0.05	0.958	-.9021273	.9512216
r20012	.1113508	.0694829	1.60	0.115	-.0282802	.2509819
r20013	-.7778491	.1171504	-6.64	0.000	-1.013272	-.5424266
r20014	0	(omitted)				
r20021	-.0562268	.4164922	-0.14	0.893	-.8931992	.7807456
r20022	.0512223	.0767245	0.67	0.508	-.1029613	.2054059
r20023	-.705417	.1038296	-6.79	0.000	-.9140704	-.4967636
r20024	0	(omitted)				
r20031	-.0539292	.3656493	-0.15	0.883	-.7887289	.6808705
r20032	.0223513	.079209	0.28	0.779	-.1368251	.1815278
r20033	-.6139207	.0941957	-6.52	0.000	-.803214	-.4246274
r20034	0	(omitted)				
r20041	-.1166739	.3168582	-0.37	0.714	-.7534244	.5200766
r20042	-.0571238	.0757494	-0.75	0.454	-.2093478	.0951003
r20043	-.5770167	.0883266	-6.53	0.000	-.7545157	-.3995178
r20044	0	(omitted)				
r20051	-.1179566	.2597649	-0.45	0.652	-.6399738	.4040606
r20052	-.0283003	.0860598	-0.33	0.744	-.2012439	.1446434
r20053	-.4934666	.0761714	-6.48	0.000	-.6465387	-.3403944
r20054	0	(omitted)				
r20061	-.1212647	.2125436	-0.57	0.571	-.548387	.3058577
r20062	-.0400965	.076217	-0.53	0.601	-.1932604	.1130674
r20063	-.4218767	.0664637	-6.35	0.000	-.5554405	-.2883129
r20064	0	(omitted)				
r20071	-.175662	.1640823	-1.07	0.290	-.5053978	.1540738
r20072	-.0303389	.0685235	-0.44	0.660	-.1680421	.1073642
r20073	-.22896	.0515107	-4.44	0.000	-.3324746	-.1254453
r20074	0	(omitted)				

r20081		-.0883956	.1177948	-0.75	0.457	-.325113	.1483218
r20082		.0144664	.0715169	0.20	0.841	-.1292521	.1581849
r20083		-.1562348	.0549373	-2.84	0.006	-.2666354	-.0458342
r20084		0	(omitted)				
r20091		-.1564882	.0699626	-2.24	0.030	-.2970834	-.0158931
r20092		-.0250962	.0628484	-0.40	0.691	-.1513947	.1012023
r20093		-.0734065	.0460022	-1.60	0.117	-.1658514	.0190384
r20094		0	(omitted)				
r20101		0	(omitted)				
r20102		0	(omitted)				
r20103		0	(omitted)				
r20104		0	(omitted)				
l_police		.1083243	.0557582	1.94	0.058	-.0037259	.2203746
unemployrt		.0131937	.0120787	1.09	0.280	-.0110794	.0374668
poverty		-.0191538	.0127408	-1.50	0.139	-.0447575	.0064498
l_income		-.3124133	.1614044	-1.94	0.059	-.6367676	.0119409
l_prisoner		-.1297827	.1936936	-0.67	0.506	-.5190245	.2594591
l_lagprisoner		-.4232017	.2471395	-1.71	0.093	-.9198471	.0734438
blackm_15_24		.0613503	.1072938	0.57	0.570	-.1542646	.2769653
whitem_15_24		.0360048	.0271789	1.32	0.191	-.0186132	.0906227
blackm_25_44		.2225389	.1231065	1.81	0.077	-.0248529	.4699308
whitem_25_44		-.0277861	.0119219	-2.33	0.024	-.0517441	-.0038281
l_exp_subsidy		-.0412795	.0390496	-1.06	0.296	-.1197527	.0371937
l_exp_pubwelf~e		.0368399	.0615966	0.60	0.553	-.0869432	.160623
trend_1		-.0927506	.012828	-7.23	0.000	-.1185295	-.0669717
trend_2		-.0319785	.0076138	-4.20	0.000	-.047279	-.016678
trend_3		-.0061905	.0082194	-0.75	0.455	-.0227081	.0103271
trend_4		-.0679086	.0118405	-5.74	0.000	-.091703	-.0441142
trend_5		-.0191227	.0064486	-2.97	0.005	-.0320816	-.0061638
trend_6		-.0023247	.0058618	-0.40	0.693	-.0141045	.009455
trend_7		.0324795	.0493962	0.66	0.514	-.066786	.1317449
trend_8		-.0173139	.0200248	-0.86	0.391	-.0575552	.0229274
trend_9		0	(omitted)				
trend_10		-.0664576	.0098661	-6.74	0.000	-.0862842	-.046631
trend_11		-.110019	.0136859	-8.04	0.000	-.1375219	-.0825161
trend_12		-.0317464	.0056227	-5.65	0.000	-.0430456	-.0204472
trend_13		.0706104	.0584627	1.21	0.233	-.0468748	.1880957
trend_14		-.0147762	.0033377	-4.43	0.000	-.0214836	-.0080688
trend_15		.0098776	.0108002	0.91	0.365	-.0118263	.0315815
trend_16		.024167	.0030567	7.91	0.000	.0180244	.0303096
trend_17		-.0028644	.0058458	-0.49	0.626	-.0146121	.0088832
trend_18		-.0512353	.0091475	-5.60	0.000	-.0696179	-.0328527
trend_19		-.0816524	.0128146	-6.37	0.000	-.1074044	-.0559005
trend_20		.08712	.0239427	3.64	0.001	.0390053	.1352346
trend_21		-.1021099	.0201221	-5.07	0.000	-.1425469	-.061673
trend_22		.0530734	.0440093	1.21	0.234	-.0353666	.1415134

trend_23		-.004965	.0089999	-0.55	0.584	-.023051	.0131211
trend_24		.0035414	.0107935	0.33	0.744	-.018149	.0252318
trend_25		-.1235853	.0149779	-8.25	0.000	-.1536846	-.093486
trend_26		.028064	.0049028	5.72	0.000	.0182114	.0379166
trend_27		.0177659	.0048846	3.64	0.001	.0079498	.0275819
trend_28		.0247214	.0044727	5.53	0.000	.0157332	.0337097
trend_29		-.0226731	.0066123	-3.43	0.001	-.035961	-.0093852
trend_30		-.0315642	.02309	-1.37	0.178	-.0779653	.0148369
trend_31		.0201153	.053031	0.38	0.706	-.0864544	.1266851
trend_32		.0696669	.0570313	1.22	0.228	-.0449417	.1842756
trend_33		-.0144914	.050741	-0.29	0.776	-.1164592	.0874764
trend_34		-.0930385	.0124652	-7.46	0.000	-.1180882	-.0679888
trend_35		.0741384	.0510922	1.45	0.153	-.0285353	.1768121
trend_36		.0379958	.0058049	6.55	0.000	.0263304	.0496611
trend_37		-.064672	.0136439	-4.74	0.000	-.0920903	-.0372536
trend_38		.0150596	.0123384	1.22	0.228	-.0097354	.0398546
trend_39		.045194	.0527444	0.86	0.396	-.0607998	.1511879
trend_40		-.0233748	.0396886	-0.59	0.559	-.103132	.0563824
trend_41		-.1005557	.0161579	-6.22	0.000	-.1330262	-.0680852
trend_42		.1814604	.0205636	8.82	0.000	.1401363	.2227845
trend_43		-.0967198	.0152166	-6.36	0.000	-.1272987	-.066141
trend_44		-.1037729	.0178143	-5.83	0.000	-.1395721	-.0679736
trend_45		-.0210785	.0055264	-3.81	0.000	-.0321841	-.0099728
trend_46		0	(omitted)				
trend_47		-.0872586	.0128651	-6.78	0.000	-.113112	-.0614052
trend_48		.008802	.0059139	1.49	0.143	-.0030824	.0206865
trend_49		0	(omitted)				
trend_50		0	(omitted)				
trend_51		0	(omitted)				
post		.076949	.0339377	2.27	0.028	.0087486	.1451494
_cons		7.798651	2.359631	3.31	0.002	3.056795	12.54051

sigma_u		2.4216987					
sigma_e		.09162631					
rho		.99857052	(fraction of variance due to u_i)				

We can also use *xtdidregress*, which gives a concise estimate of the *ATE* and allows for us to test for pre-treatment trends.

```
xtdidregress (l_homicide i.year $region $xvar $lintrend) (post) [aweight=popwt], group
```

note: r20004 omitted because of collinearity.
note: r20014 omitted because of collinearity.
note: r20024 omitted because of collinearity.

note: r20034 omitted because of collinearity.
 note: r20044 omitted because of collinearity.
 note: r20054 omitted because of collinearity.
 note: r20064 omitted because of collinearity.
 note: r20074 omitted because of collinearity.
 note: r20084 omitted because of collinearity.
 note: r20094 omitted because of collinearity.
 note: r20101 omitted because of collinearity.
 note: r20102 omitted because of collinearity.
 note: r20103 omitted because of collinearity.
 note: r20104 omitted because of collinearity.
 note: trend_9 omitted because of collinearity.
 note: trend_46 omitted because of collinearity.
 note: trend_49 omitted because of collinearity.
 note: trend_50 omitted because of collinearity.
 note: trend_51 omitted because of collinearity.

Treatment and time information

Time variable: year

Control: post = 0

Treatment: post = 1

```

-----
                |   Control   Treatment
-----+-----
Group          |
      sid      |           29           21
-----+-----
Time           |
  Minimum      |           2000           2006
  Maximum      |           2000           2010
-----

```

Difference-in-differences regression

Number of obs = 550

Data type: Longitudinal

(Std. err. adjusted for 50 clusters in sid)

```

-----
                |               Robust
      l_homicide | Coefficient  std. err.      t    P>|t|    [95% conf. interval]
-----+-----
ATET            |
      post      |
  (1 vs 0)      |    .076949   .0339377     2.27   0.028    .0087486    .1451494
-----

```

Note: ATET estimate adjusted for covariates, panel effects, and time effects.

Note: Treatment occurs at different times.

From our TWFE, we find that the *ATE* is equal to 0.076949. Castle-doctrine statutes increase homicides per 100K by $(e^{0.076949} - 1) * 100\% = 8.0\%$, which is statistically significant at the 5% level.

1.4 Event Study

```
xtevent l_homicide $region [aweight=popwt], pol(post) cluster(sid) timevar(year) window
```

Using options panelvar and timevar from xtset

No proxy or instruments provided. Implementing OLS estimator
The calculated window by window(max) is (-9,3), plus the endpoints -10 and 4.
(converged in 3 iterations)

```
note: r20002 omitted because of collinearity
note: r20013 omitted because of collinearity
note: r20024 omitted because of collinearity
note: r20032 omitted because of collinearity
note: r20034 omitted because of collinearity
note: r20043 omitted because of collinearity
note: r20054 omitted because of collinearity
note: r20062 omitted because of collinearity
note: r20072 omitted because of collinearity
note: r20083 omitted because of collinearity
note: r20084 omitted because of collinearity
note: r20092 omitted because of collinearity
note: r20101 omitted because of collinearity
note: r20102 omitted because of collinearity
```

HDFE Linear regression	Number of obs	=	550
Absorbing 2 HDFE groups	F(44, 49)	=	295.06
Statistics robust to heteroskedasticity	Prob > F	=	0.0000
	R-squared	=	0.9444
	Adj R-squared	=	0.9315
	Within R-sq.	=	0.1807
Number of clusters (sid)	=	50	Root MSE = 0.1083

(Std. err. adjusted for 50 clusters in sid)

```
-----+-----
          |               Robust
l_homicide | Coefficient  std. err.      t    P>|t|    [95% conf. interval]
```


_k_eq_m10		-.2670587	.0464639	-5.75	0.000	-.3604314	-.173686
_k_eq_m9		-.3136291	.0810769	-3.87	0.000	-.4765593	-.150699
_k_eq_m8		-.1480105	.0852785	-1.74	0.089	-.3193841	.0233631
_k_eq_m7		-.0011922	.0574174	-0.02	0.984	-.1165768	.1141923
_k_eq_m6		-.0033678	.0473356	-0.07	0.944	-.0984923	.0917567
_k_eq_m5		-.0187262	.0486225	-0.39	0.702	-.1164367	.0789844
_k_eq_m4		.0085331	.0356398	0.24	0.812	-.0630878	.080154
_k_eq_m3		.0049404	.0339288	0.15	0.885	-.0632421	.0731222
_k_eq_m2		-.0322577	.034501	-0.93	0.354	-.10159	.0370745
_k_eq_p0		.0758257	.0281218	2.70	0.010	.0193127	.1323386
_k_eq_p1		.0648564	.0475352	1.36	0.179	-.0306691	.1603819
_k_eq_p2		.0905699	.0586977	1.54	0.129	-.0273875	.2085274
_k_eq_p3		.0918709	.0565459	1.62	0.111	-.0217624	.2055042
_k_eq_p4		.1844643	.056076	3.29	0.002	.0717754	.2971532
r20001		-.3176814	.1055113	-3.01	0.004	-.5297144	-.1056485
r20002		0	(omitted)				
r20003		-.1014046	.0678023	-1.50	0.141	-.2376584	.0348492
r20004		-.1735281	.0529523	-3.28	0.002	-.2799397	-.0671165
r20011		-.097969	.0996036	-0.98	0.330	-.2981299	.102192
r20012		.136426	.0652085	2.09	0.042	.0053847	.2674674
r20013		0	(omitted)				
r20014		.0644022	.0543544	1.18	0.242	-.0448271	.1736314
r20021		-.2108039	.0728759	-2.89	0.006	-.3572534	-.0643543
r20022		.0004829	.048146	0.01	0.992	-.0962701	.097236
r20023		-.0783909	.046152	-1.70	0.096	-.1711368	.014355
r20024		0	(omitted)				
r20031		-.1710896	.079739	-2.15	0.037	-.3313312	-.010848
r20032		0	(omitted)				
r20033		-.0556467	.0549892	-1.01	0.317	-.1661516	.0548581
r20034		0	(omitted)				
r20041		-.08143	.0741366	-1.10	0.277	-.230413	.067553
r20042		.0413541	.0596014	0.69	0.491	-.0784194	.1611277
r20043		0	(omitted)				
r20044		.1008408	.0384528	2.62	0.012	.023567	.1781145
r20051		-.1464866	.0682065	-2.15	0.037	-.2835527	-.0094205
r20052		-.002504	.0601266	-0.04	0.967	-.1233329	.1183249
r20053		-.0910554	.0515309	-1.77	0.083	-.1946106	.0124998
r20054		0	(omitted)				
r20061		-.1217555	.0545111	-2.23	0.030	-.2312997	-.0122112
r20062		0	(omitted)				
r20063		-.0808163	.0693046	-1.17	0.249	-.2200891	.0584566
r20064		-.0086811	.0605718	-0.14	0.887	-.1304046	.1130425
r20071		-.1647261	.0589828	-2.79	0.007	-.2832565	-.0461957
r20072		0	(omitted)				
r20073		-.0110104	.0615684	-0.18	0.859	-.1347367	.1127159

r20074		-.0173833	.0625427	-0.28	0.782	-.1430676	.1083011
r20081		-.0441083	.0705833	-0.62	0.535	-.1859507	.0977342
r20082		.0834497	.0698646	1.19	0.238	-.0569485	.2238478
r20083		0	(omitted)				
r20084		0	(omitted)				
r20091		-.1250185	.0483505	-2.59	0.013	-.2221826	-.0278545
r20092		0	(omitted)				
r20093		-.059229	.0684898	-0.86	0.391	-.1968645	.0784065
r20094		-.0440906	.0717044	-0.61	0.541	-.188186	.1000047
r20101		0	(omitted)				
r20102		0	(omitted)				
r20103		-.1376362	.0624248	-2.20	0.032	-.2630836	-.0121889
r20104		-.1100676	.06304	-1.75	0.087	-.2367512	.016616

Absorbed degrees of freedom:

Absorbed FE	Num. Coefs.	=	Categories	-	Redundant
sid	0		50		50 *
year	10		11		1

* = fixed effect nested within cluster; treated as redundant for DoF computation

Next, we will plot our leads and lags of the event study.

```
xteventplot, xtitle("Years from policy") ytitle("LN of Homicide Rate per 100K") title("Homicide Rate per 100K")
```

Except for year 9 and year 8, there is no statistically significant difference between treatment and never treated. There may be spurious factors influencing year 9 and year 8 before the policy and there are very few units in the lead9 (1 out of 550) and lead8 (3 out of 550), so it is probably best to disregard.

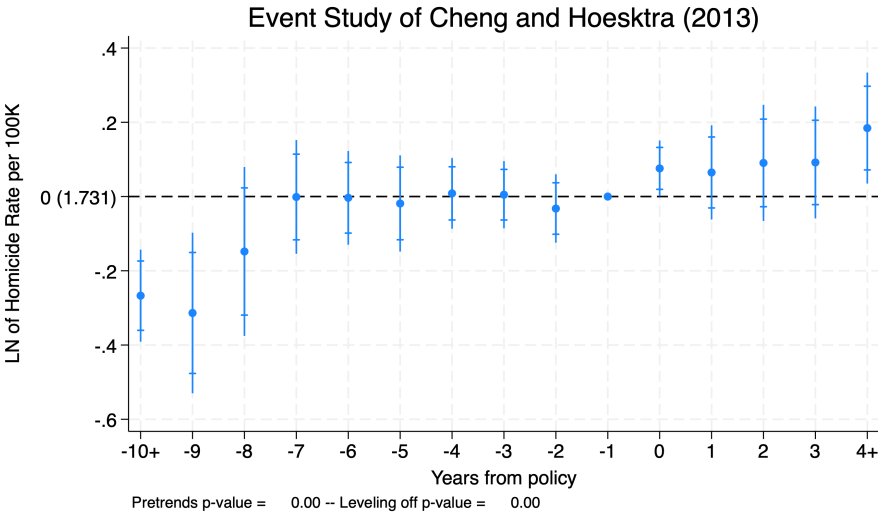


Figure 1.3: Event Study of Cheng and Hoesktra (2013)

Chapter 2

Goodman-Bacon Decomposition

We will again use the data from Cheng and Hoesktra (2013) to assess the Bacon-Goodman Decomposition of the different *ATE*s when there is differential timing. We have a few options to calculate the individual *ATE*s and different variance weights. We will focus on *estat bdecomp*, *bacondecomp* and *ddtiming*. Please note that *ddtiming* has been depreciated.

2.1 *estat bdecomp*

Our first method to implement the Bacon-Goodman Decomposition is *estat bdecomp*. We run this command after we use *xtdidregress* or *didregress*. Please note that we use the option *graph* to display the individual *ATE* and the corresponding weights.

```
xtdidregress (l_homicide) (post), group(sid) time(year)
estat bdecomp, graph
```

Treatment and time information

Time variable: year

Control: post = 0

Treatment: post = 1

```
-----
| Control Treatment
-----+-----
Group |
```

	sid	29	21
-----+			
Time			
Minimum		2000	2006
Maximum		2000	2010

Difference-in-differences regression
Data type: Longitudinal

Number of obs = 550

(Std. err. adjusted for 50 clusters in sid)

-----+-----						
		Robust				
l_homicide		Coefficient	std. err.	t	P> t	[95% conf. interval]
-----+-----						
ATET						
post						
(1 vs 0)		.0693984	.0558596	1.24	0.220	-.0428557 .1816526

Note: ATET estimate adjusted for panel effects and time effects.

Note: Treatment occurs at different times.

DID treatment-effect decomposition

ATET = .0693984

Number of obs = 550

Number of groups = 50

Number of cohorts = 6

ATET decomposition summary	ATET component	Weight

Treated vs never treated	.07843799	0.898809
Treated earlier vs later	-.02857716	0.077079
Treated later vs earlier	.04563468	0.024112

Full ATET decomposition	2x2 coefficient	Weight

Treated vs never treated		
2006 vs never treated	.14503261	0.050306
2007 vs never treated	.05925429	0.610385
2008 vs never treated	.09201016	0.160981
2009 vs never treated	.18195417	0.060368
2010 vs never treated	.07398961	0.016769

Treated earlier vs later

2006 vs 2007	.04200339	0.004510
2006 vs 2008	.09126019	0.002776
2006 vs 2009	.05519713	0.002082
2006 vs 2010	-.05417012	0.001388
2007 vs 2008	-.0219597	0.021048
2007 vs 2009	.01509344	0.021048
2007 vs 2010	-.07101927	0.015786
2008 vs 2009	-.15384871	0.003701
2008 vs 2010	-.21834129	0.003701
2009 vs 2010	-.04052124	0.001041
Treated later vs earlier		
2007 vs 2006	-.04349169	0.003007
2008 vs 2006	.05369758	0.001388
2009 vs 2006	.13322861	0.000694
2010 vs 2006	-.01935504	0.000231
2008 vs 2007	.04379672	0.009020
2009 vs 2007	.14954754	0.006014
2010 vs 2007	.00654799	0.002255
2009 vs 2008	-.14971815	0.000925
2010 vs 2008	-.21479646	0.000463
2010 vs 2009	-.02278972	0.000116

Note: Number of cohorts includes never treated.

Note: The ATET reported by `xtddidregress` is a weighted average of the ATET components. If any component is substantially different from the ATET reported by `xtddidregress` and the weight is large, consider accounting for treatment-effect heterogeneity by using `xthddidregress`.

We can check the weights of the individual *ATET* to examine the heterogeneity of the estimates. The summary shows the Treated vs Never Treated, Treated Earlier vs Treated Later, and Treated Later vs Treated Earlier *ATETs* along with their corresponding weights.

We can see that Treated vs Never Treated has the largest weight of 89.8%, from our graph and our summary table.

We can delve into the early treated vs late treated in the cohort full decomposition. An interesting outcome is that our Treated vs Never Treated *ATET* are always positive and range from 0.059 to 0.182.

However, our Early Treated vs Late Treated and Late Treated vs Early Treated have a lot of heterogeneity. 9 are positive, while 11 are negative. It indicated we may need to worry about heterogeneity bias. However, the weights for both cohorts is 10.1% (7.7% for early treated vs late treated and 2.4% for late treated vs early treated).

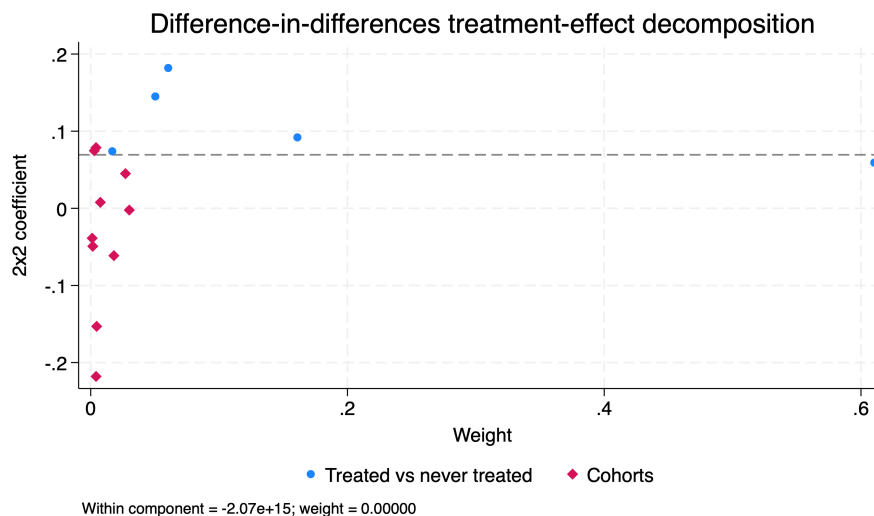


Figure 2.1: Bacon-Goodman Decomposition

2.2 *bacondecomp*

Our second method to implement the Bacon-Goodman Decomposition is *bacondecomp*

```
ssc install bacondecomp
```

We will first run a TWFE regression with *areg* and then run *Bacondecomp* after regression, but it generates stubs. Notice that the *areg* will have similar outcome as our *xtdidregress*

```
areg l_homicide post i.year, absorb(sid) robust
bacondecomp l_homicide post, stub(Bacon_) robust ddetail
drop Bacon_*
```

Linear regression, absorbing indicators
Absorbed variable: sid

Number of obs	=	550
No. of categories	=	50
F(11, 489)	=	4.73
Prob > F	=	0.0000
R-squared	=	0.9102
Adj R-squared	=	0.8991
Root MSE	=	0.1874

l_homicide	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
post	.0693984	.034312	2.02	0.044	.0019813	.1368155
year						
2001	.0234081	.0481165	0.49	0.627	-.0711326	.1179488
2002	.002241	.0437824	0.05	0.959	-.0837839	.0882659
2003	.0476296	.0428512	1.11	0.267	-.0365656	.1318247
2004	.04259	.0399801	1.07	0.287	-.035964	.1211441
2005	.0609827	.0413633	1.47	0.141	-.0202889	.1422544
2006	.0756094	.0401986	1.88	0.061	-.0033739	.1545927
2007	.0614879	.0467341	1.32	0.189	-.0303365	.1533123
2008	.0125426	.0479081	0.26	0.794	-.0815885	.1066738
2009	-.0690221	.0460227	-1.50	0.134	-.1594487	.0214046
2010	-.1271772	.0460306	-2.76	0.006	-.2176194	-.036735
_cons	1.384578	.0358783	38.59	0.000	1.314084	1.455073

Computing decomposition across 6 timing groups
including a never-treated group

l_homicide	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
post	.0693984	.0558596	1.24	0.214	-.0400844	.1788813

Bacon Decomposition

	Beta	TotalWeight
Early_v_Late	.0420033932	.0045102348
Late_v_Early	-.0434916914	.0030068233
Early_v_Late	.0912601873	.0027755292
Late_v_Early	.0536975749	.0013877646
Early_v_Late	-.021959696	.0210477626
Late_v_Early	.0437967181	.0090204695
Early_v_Late	.0551971309	.0020816468
Late_v_Early	.1332286149	.0006938823
Early_v_Late	.0150934393	.0210477626
Late_v_Early	.1495475471	.0060136465
Early_v_Late	-.1538487077	.0037007056
Late_v_Early	-.1497181505	.0009251764
Early_v_Late	-.0541701168	.0013877646

```

|           Late_v_Early | -.0193550438   .0002312941 |
|           Early_v_Late | -.0710192695   .0157858219 |
|           Late_v_Early | .0065479889    .0022551174 |
|           Early_v_Late | -.218341291    .0037007056 |
|           Late_v_Early | -.2147964537   .0004625882 |
|           Early_v_Late | -.040521238    .0010408234 |
|           Late_v_Early | -.0227897167   .000115647 |
|           Never_v_timing | .0784379909    .8988088336 |
+-----+

```

Notice that the TWFE estimator is the same between *areg* and *xtddidregress*. One thing about this command is that we lack details of the individual groups. We only see Early_v_Late and Late_v_Early. I personally prefer the *estat bdecomp* command.

2.3 *ddtiming*

```
net install ddtiming, from(https://raw.githubusercontent.com/tgoldring/ddtiming/master)
```

```
ddtiming l_homicide post, i(sid) t(year) yline(0)
```

```

Calculating treatment times...
Calculating weights...
Estimating 2x2 diff-in-diff regressions...

```

```
Diff-in-diff estimate: 0.069
```

DD Comparison	Weight	Avg DD Est
Earlier T vs. Later C	0.077	-0.029
Later T vs. Earlier C	0.024	0.046
T vs. Never treated	0.899	0.078

```
T = Treatment; C = Comparison
```

We get similar results, but the issue is that *ddtiming* has been depreciated.

Cheng, Cheng, and Mark Hoekstra. 2013. “Does Strengthening Self-Defense Law Deter Crime or Escalate Violence? Evidence from Expansions to Castle Doctrine.” *Journal of Human Resources* 48 (3): 821–54.

Cunningham, S. (2021). *Causal Inference: The Mixtape*. Yale University Press.

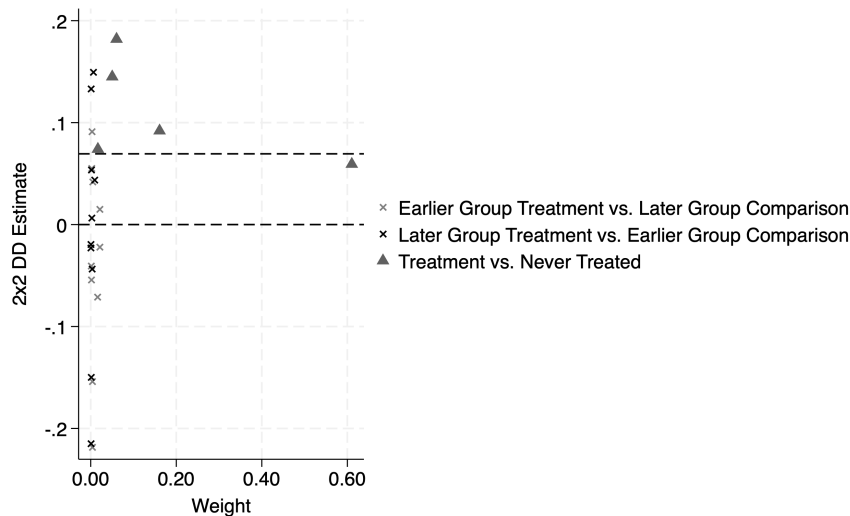


Figure 2.2: Bacon-Goodman Decomposition with *ddtiming*

Freyaldenhoven, S., Hansen C. & Perez, JEP., and Shapiro J. (2021). *XTEVENT: Stata module to estimate and visualize linear panel event-study models*. Statistical Software Components S458987, Boston College Department of Economics, revised 12 Jul 2024. <https://ideas.repec.org/c/boc/bocode/s458987.html>

Gruber, Levine, and Staiger (1999). Gruber, J., Levine, P.B., and Staiger, D. (1999). Abortion Legalization and Child Living Circumstances: Who Is the 'Marginal Child'? *The Quarterly Journal of Economics* 114 (1): 263–91.

Huber, M. (2023). *Causal Analysis: Impact Evaluation and Causal Machine Learning with Applications in R*. The MIT Press.

Huntington-Klein, N. (2022). *The Effect: An Introduction to Research Design and Causality*. CRC Press.

Levitt, S.D. (2004). Why Crime Fell in the 1990s: Four Factors That Explain the Decline and Six That Do Not. *Journal of Economic Perspectives* 18 (1): 163-190

Princeton (2026). *Difference-in-Differences in Stata: A step-by-step guide*. <https://libguides.princeton.edu/stata-did/1#s-lg-box-wrapper-41689126>